

"Modeling information asymmetry in financial markets using game-theoretic signaling frameworks — can NLP analysis of earnings call transcripts detect when insiders are signaling private information, and does market behavior follow predicted Nash Equilibrium responses?"

The Real World Situation

Every quarter, companies hold **earnings calls**. The CEO and CFO get on a call with analysts and investors and talk about how the company performed and what they expect going forward. These calls are public — anyone can read the transcript.

Now here's the interesting thing: **management knows way more than investors do**. The CFO knows if next quarter is going to be rough. The CEO knows if a big deal is about to close. Investors don't know any of this. This gap is called **information asymmetry** — one side of the table has more information than the other.

But here's what's fascinating: even though management can't directly say "sell our stock, things are bad" (that's illegal), **they communicate through language**. Vague answers, hedging words like "we remain cautiously optimistic," unusually specific numbers when things are good, avoiding certain topics — all of this is a form of signaling. And savvy investors try to decode these signals.

This is the game.

The Game Theory Layer

Now let's put the game theory framework on top of this.

Who are the players?

- Management (CEO/CFO) — they have private information
- Informed traders (hedge funds, institutional investors) — they're trying to decode signals
- Retail investors — largely uninformed, react to price movements

What are the strategies?

- Management chooses: how much to reveal, how to phrase things, what to emphasize
- Informed traders choose: buy, sell, or hold — based on their interpretation of signals

- Retail investors follow price momentum largely

What is a signaling game formally?

Imagine management knows the company's true health — call it either Good (G) or Bad (B). Investors can't see this directly. Management sends a "signal" through their language on the earnings call. Investors observe the signal and update their belief about whether the company is G or B, then trade accordingly. The stock price moves. Management's reputation and stock compensation depends on this outcome.

The Nash Equilibrium in this game is a situation where: management is signaling in the best way given how investors interpret signals, AND investors are interpreting signals in the best way given how management signals. Neither side wants to change their strategy.

Why is this interesting? Because in equilibrium, you can sometimes predict what the market should do after certain language patterns. If the market doesn't do what theory predicts — that's a finding worth investigating.

The ML/NLP Layer

This is where your existing skills come in directly. You'll use NLP to:

- Read thousands of earnings call transcripts
- Extract linguistic features — hedging language, sentiment, specificity, topic avoidance
- Classify whether the signal is "positive" or "negative" in game-theoretic terms
- Then check: did the stock price move the way the equilibrium prediction said it should?

Your model is essentially asking: **"Can we teach a machine to read earnings calls the way a savvy informed trader would?"**